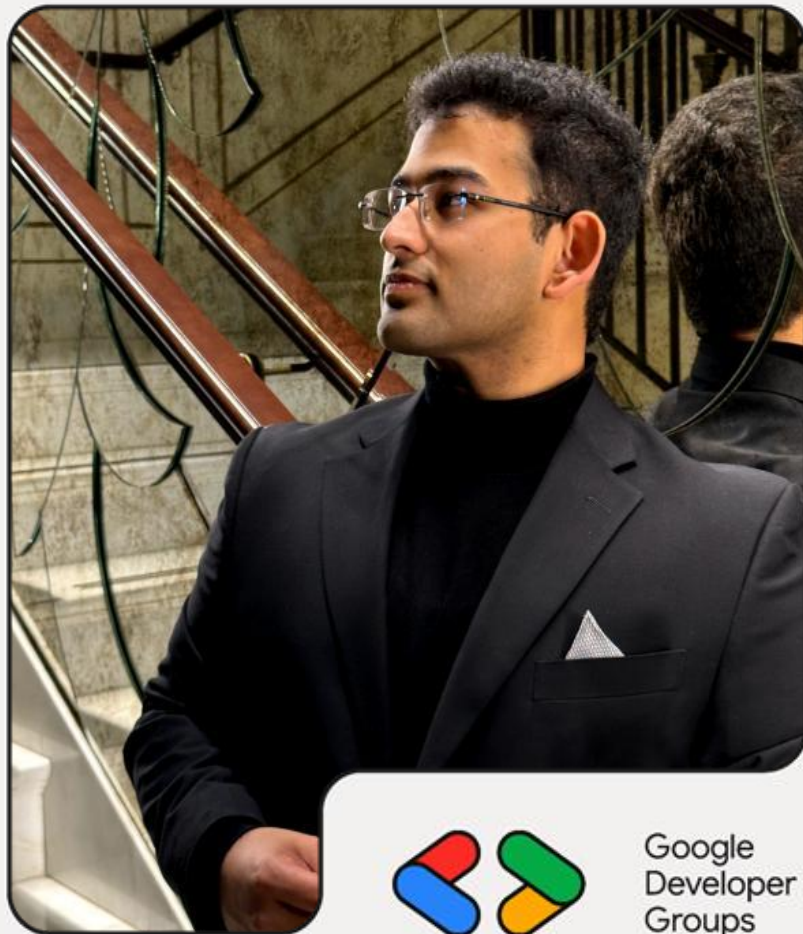




Intro To Building Agents With Google's ADK

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Google Developer Expert – Cloud AI

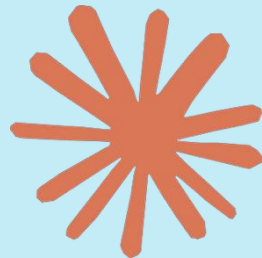


Google
Developer
Groups

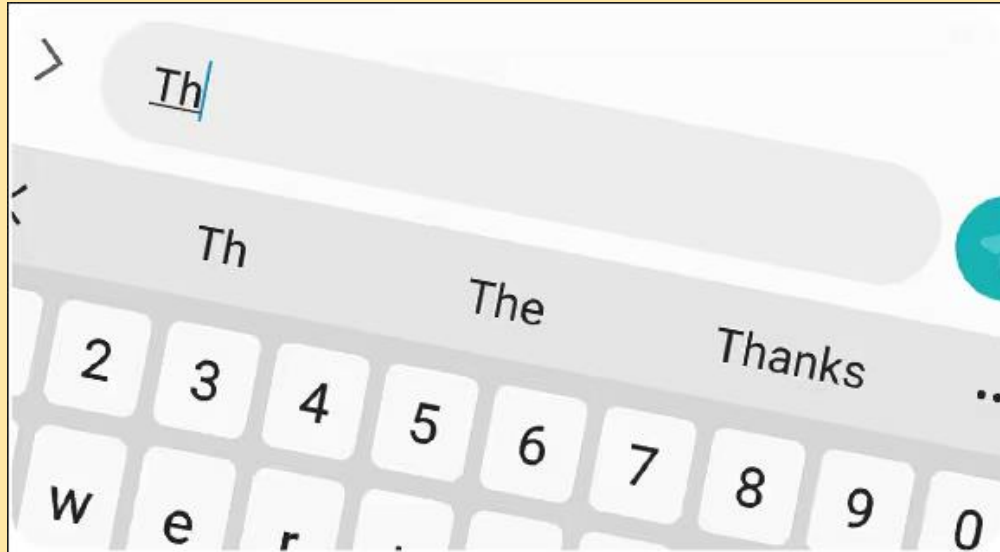
Using LLMs daily

- Write an intro for my essay
- Explain this question in the previous exam
- Give me code to solve Three Sum LeetCode problem
- **But why not:**
 - “Check my university portal for my grades, reference against my syllabus, find how out much I need to get an A, go through the slides, the web, and previous exams to create notes I can study from to guaruntee an A” ?
- LLMs are **passive**, **stateless**, and **can't interact with the outside world**

Gemini



What is an LLM?

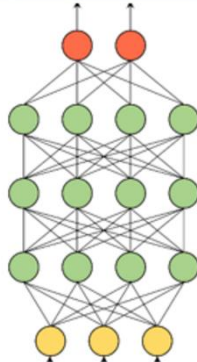


What is an LLM?

Probabilities
over
vocabulary

bread	...	cake	...	dress	...	game	...	salad
0.02	...	0.25	...	0.001	...	0.001	...	0.06

Language
Model



Input from User

Sorry I'm running

late again

Sample a token from
vocabulary

Run LM for the next token

I'm running late

Generate new input

Generated text:

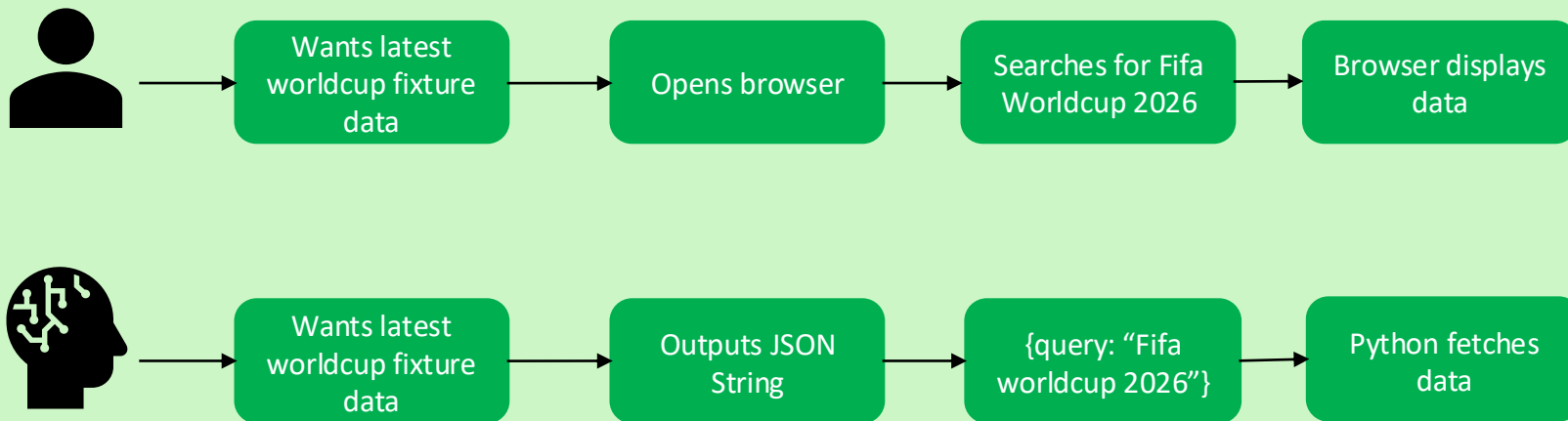
Sorry I'm running late again

What is an LLM?

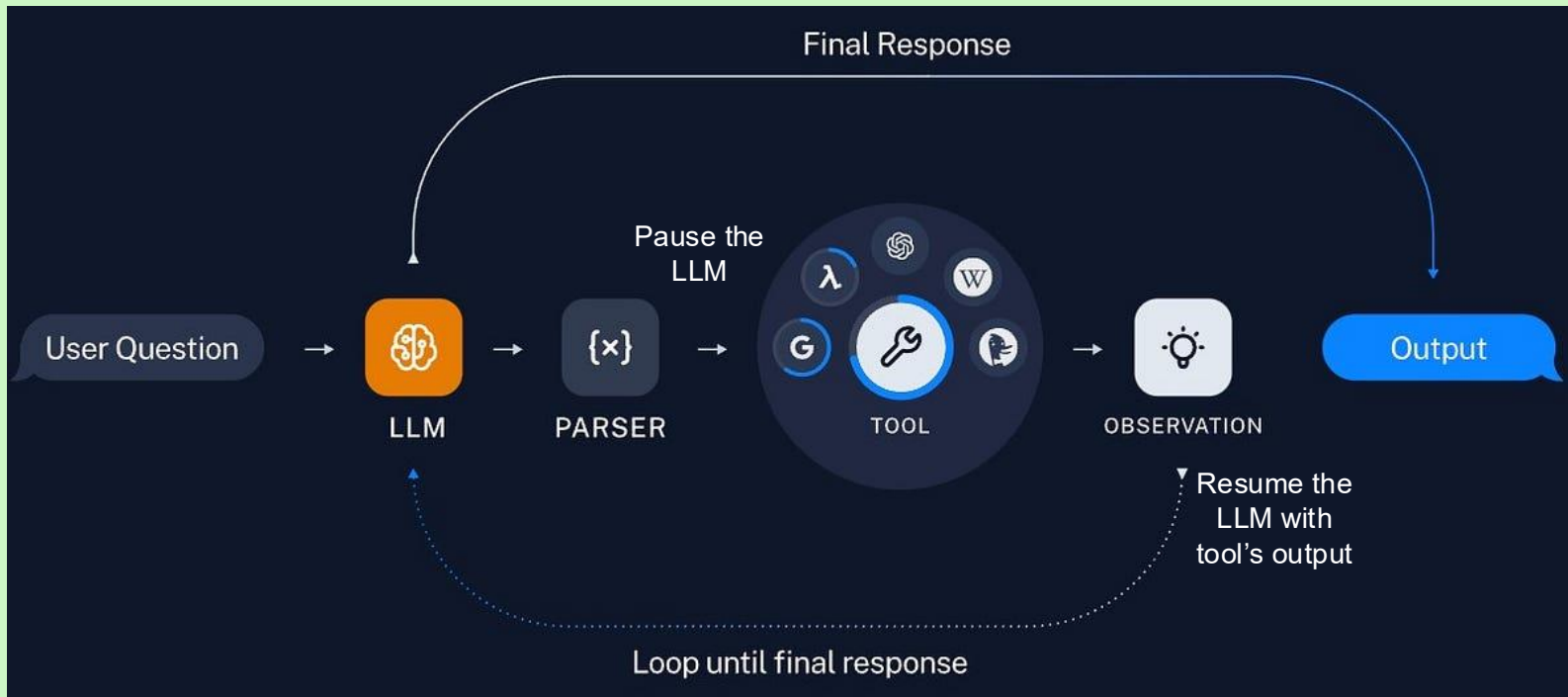
- Neural network that is REALLY good at next word prediction
- Trained on the entire internet – Huge Knowledge base
- Has become good at reasoning
- BUT... Isolated

What is a Tool call?

- The “hands” of the LLM



What is a Tool call?

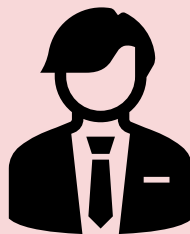


LLMs vs Agentic AI



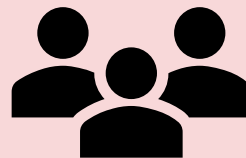
LLM

- Reasoning and knowledge, but can't take actions
- Stateless
- Passive



Agent

- Body = tools
- Job Desc = instructions & goals
- Autonomous
- Stateful
- Take actions till job is done



Multi-Agent

- Talented and focused individuals
- Get tasks to help achieve goal
- Focused
- Modular, reliable, debug-able

Why Agent Development Kit (ADK)?

- Need more control to build more complex agents and workflows
 - Modularity
 - Reusability
 - State management
- Easy to get started & easy to scale
- Helps in quick development for hackathons and competitions
- Gaining traction in industry – Companies are looking for agentic AI developers!

Case Study – Personal Investment Bot

- You want to quickly know how a certain stock is performing in a given time period
- You'll need some market research on latest news + numerical backing
- You should be able to ask for a given stock and receive a report with relevant info and suggestions
- Let's build an AI workflow with Agent Development Kit!
- (And how you can adapt this to your own projects)

Adaptable Design – Other Use Cases

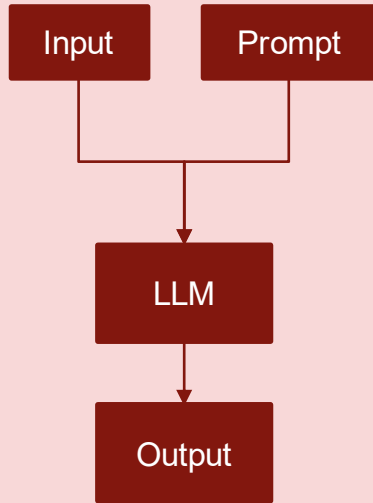
- Weekly sales reports - Sales
- Daily operations status reports - Operations
- Chatting with your data - General
- Strategic planning - Strategy
- Building a prototype in a hackathon next weekend!

Personal Investment Bot

- Take a particular stock and time period, and conduct analysis on how the stock performed
- Search the web for headlines about the stock
- Analyze the market sentiment towards the stock
- Fetch and analyze temporal data (prices & volume of the stock) in that time period
- Collate data, summarize, and identify actionable insights
- Make an organized report

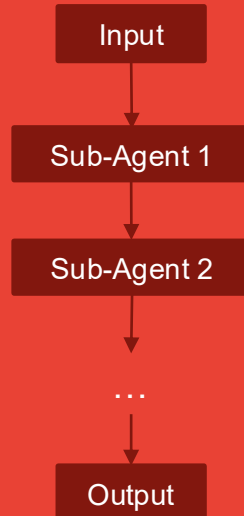
ADK Agents

LLM Agent

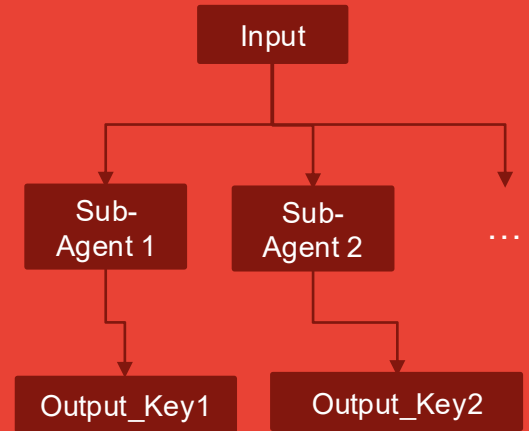


Workflow Agent

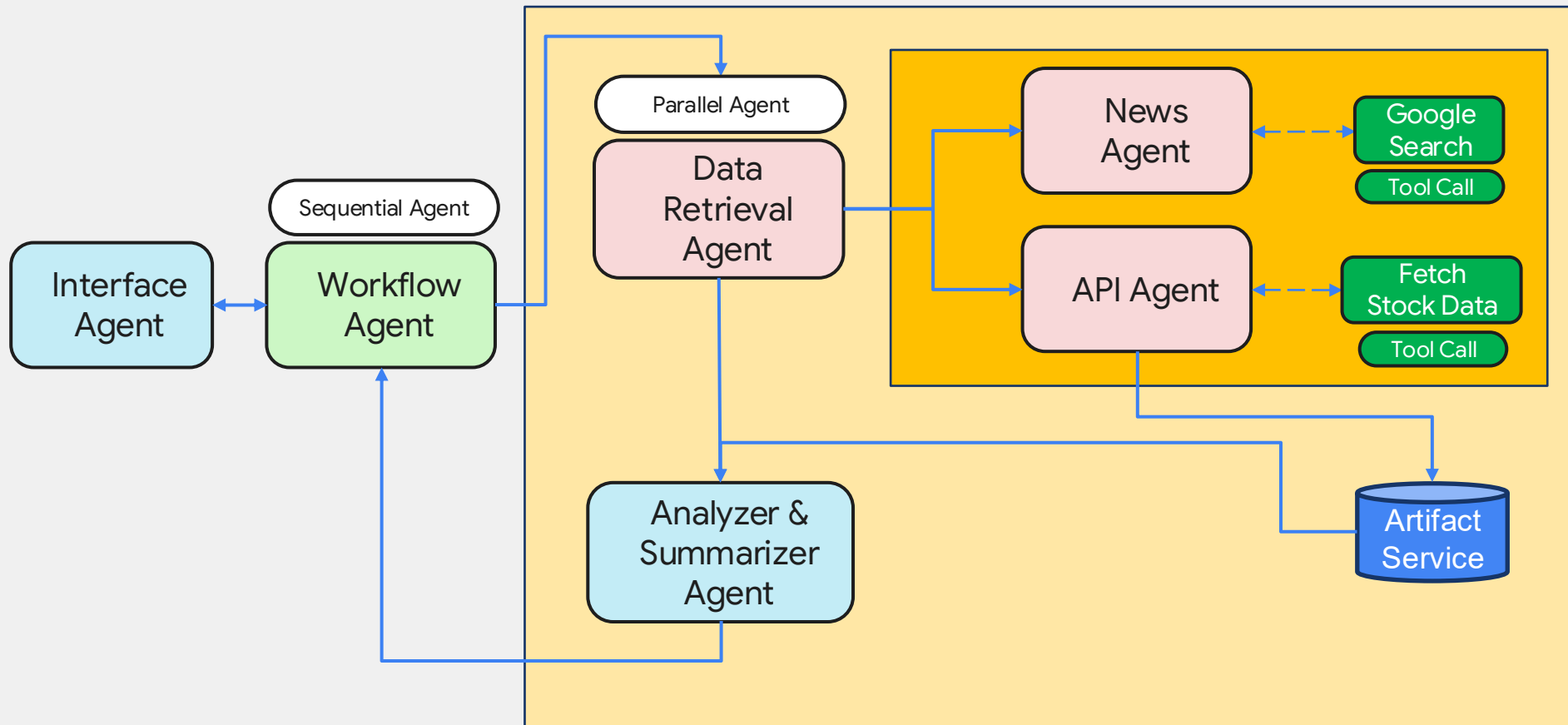
Sequential Agent



Parallel Agent



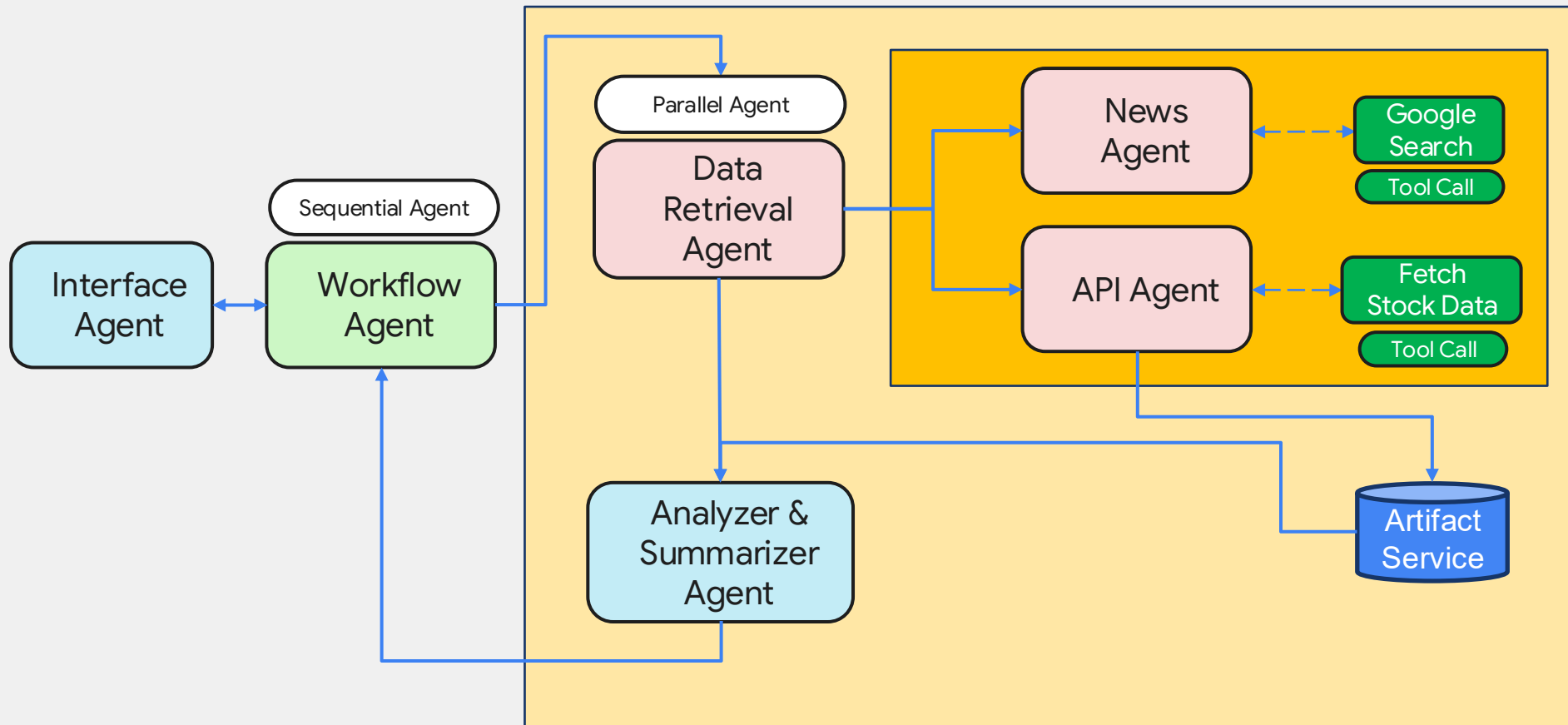
Our Architecture



Interface Agent

- Interacts with the user
- Ask the user for a stock (ticker or name of publicly listed company)
- Ask the user for a date range
- Validate inputs
- Initiate workflow
- Return result to user as message

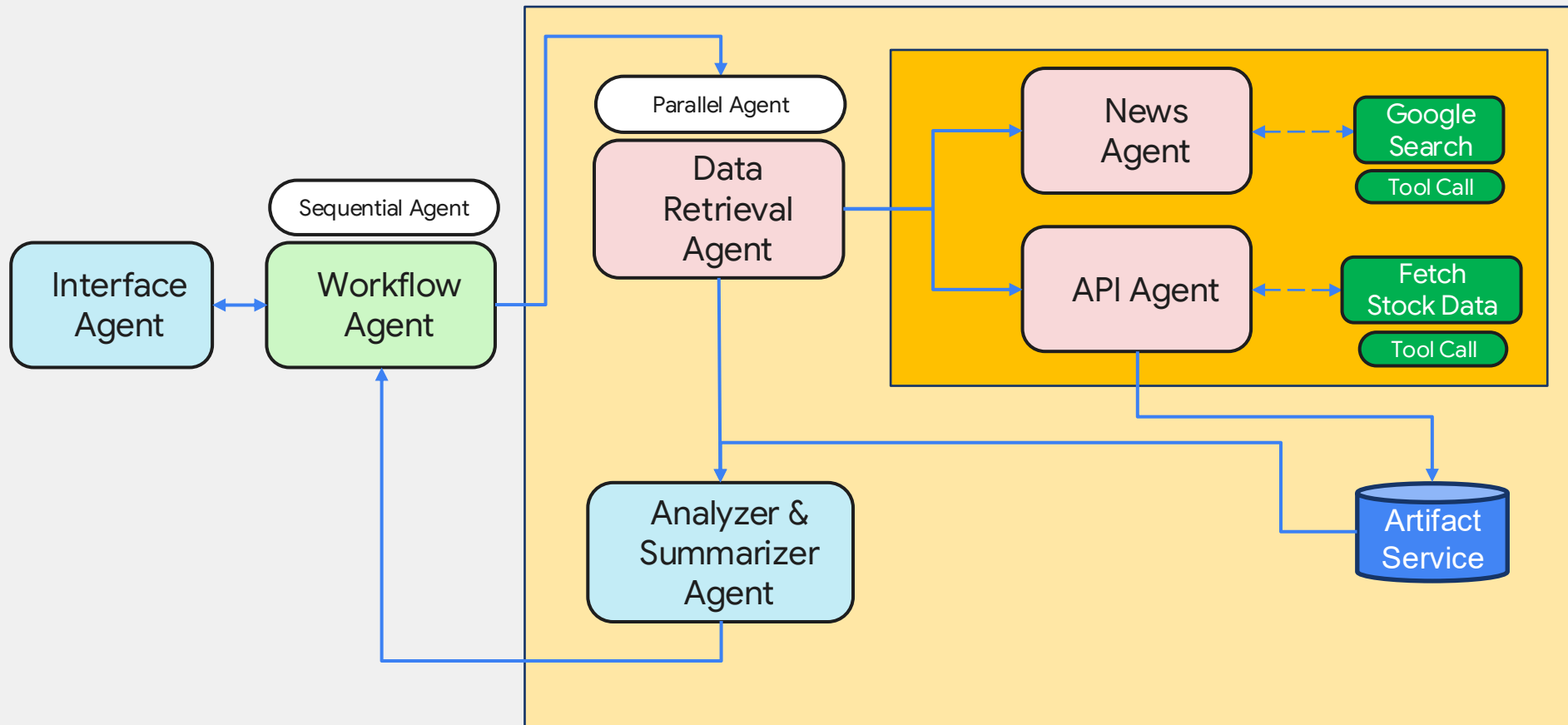
Our Architecture



Workflow Agent

- Sequential agent
- Input text from the Interface agent
 - Eg: “Analyze GOOG stock performance, news sentiment, technical indicators, and provide trading recommendations for the period 2025-08-01 to 2025-08-31.”
- First execute Data Retriever Agent (result stored in output_key)
- Then execute Analyzer & Summarizer Agent (result stored in output_key)
- Once done, return control to Interface agent

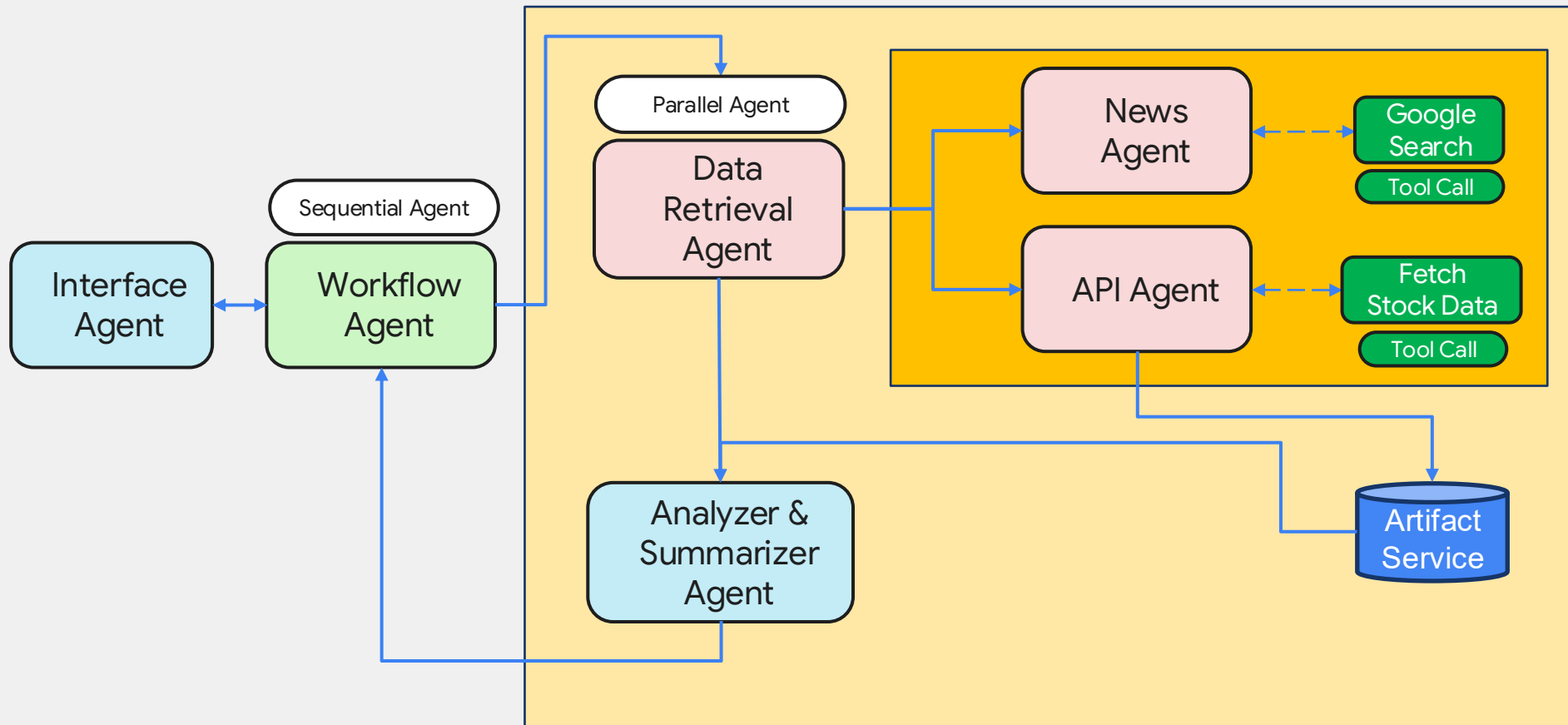
Our Architecture



Data Retrieval Agent

- Parallel agent
- Execute News Agent
- Execute API Agent
- Once done, return control to Workflow agent

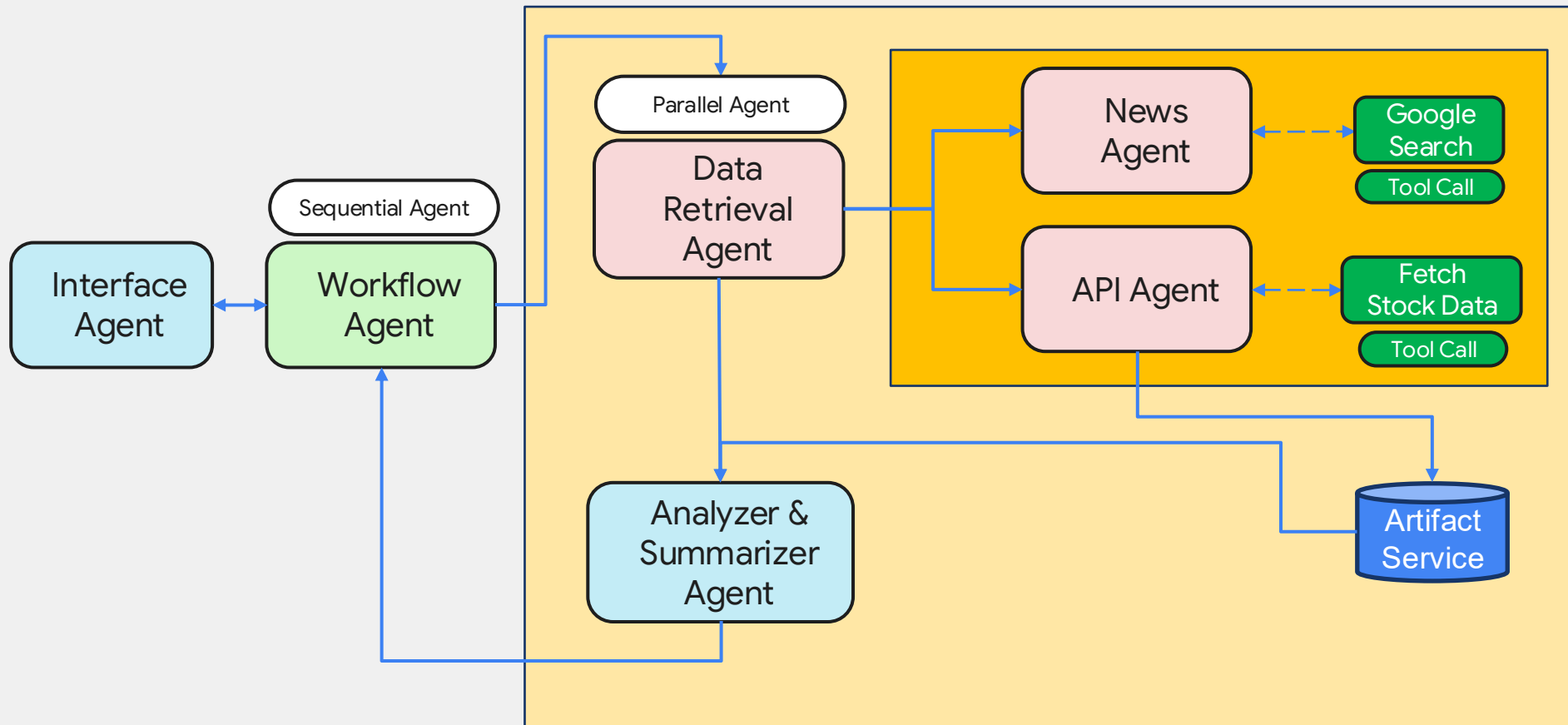
Our Architecture



News Agent

- Performs several Google searches with different search keys – Google Search tool call
- Identifies:
 - Summary of latest news articles & developments of given stock
 - Top 5 most impactful headlines
 - Market sentiment (Positive, Neutral, Negative)
- Saves the results in JSON format in an output key “news_agent_output”

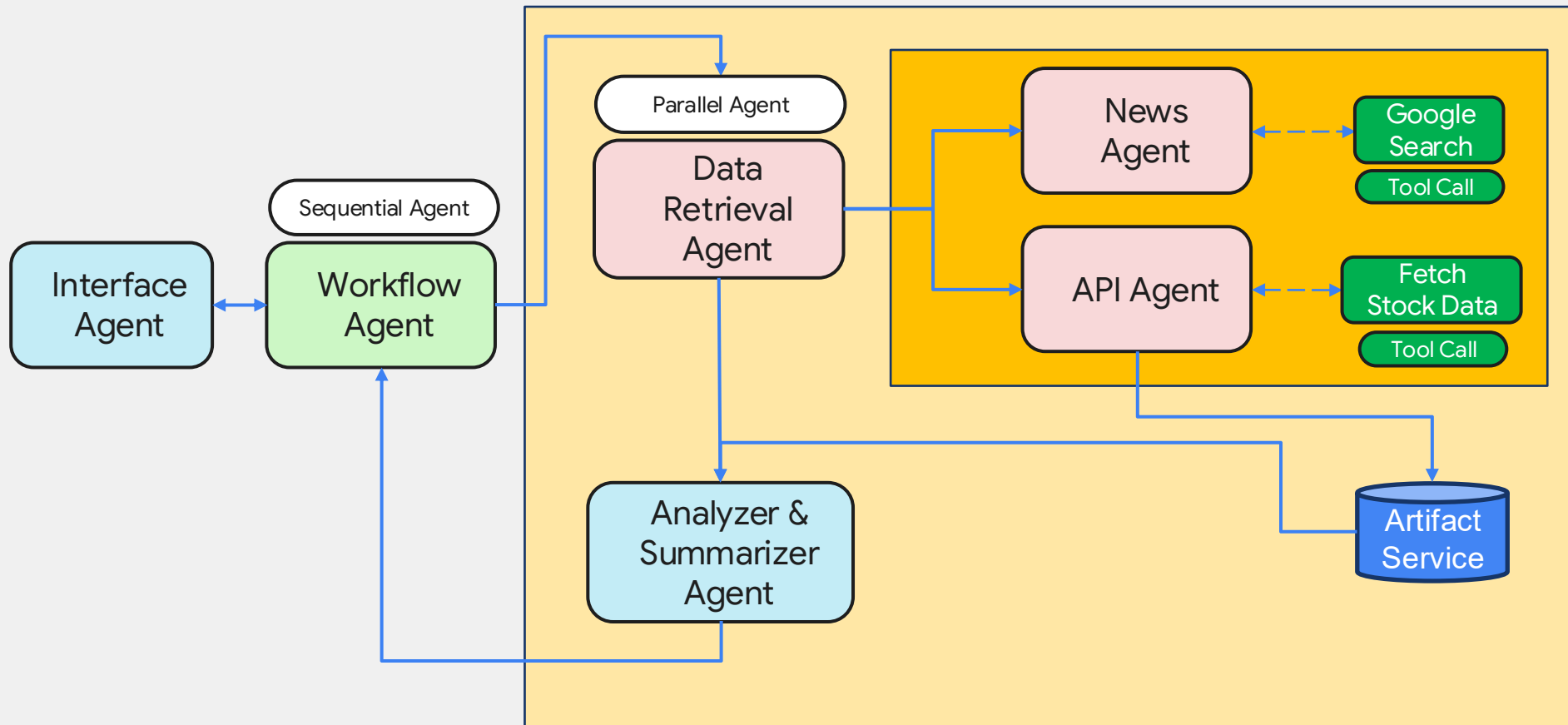
Our Architecture



API Agent

- Uses a tool-call to fetch data for stock from Yahoo finance API (yfinance)
- Downloads the open/close/high/low/volume data (as dataframe) for each date
- Saves the data as a JSON in artifact service
 - **Artifact:** Any large object that is loaded/created
 - Includes files, images, data, etc
 - **Artifact service:** A temporary storage space for the session
 - Easy way to transfer large objects between agents
 - Saves to RAM / GCS / Redis

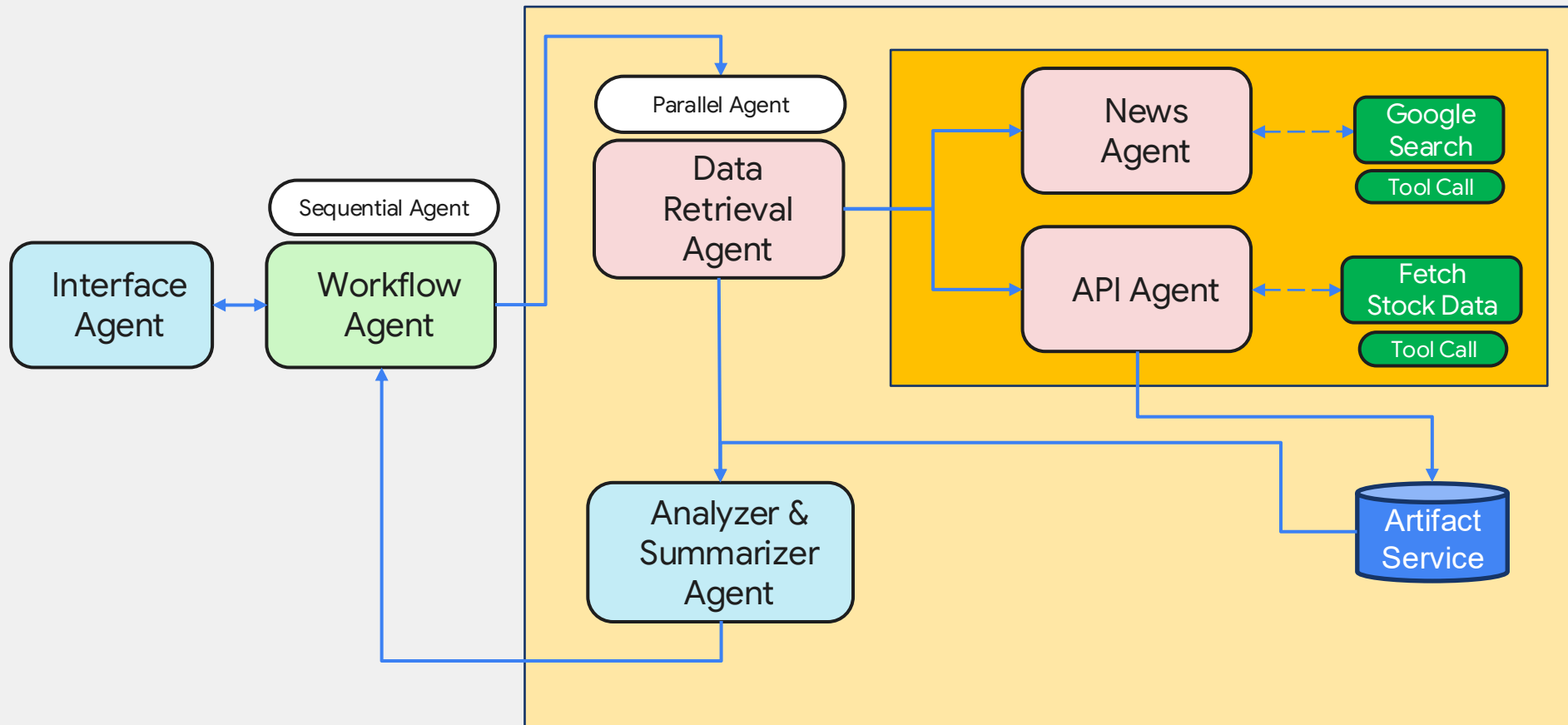
Our Architecture



Analyzer & Summarizer Agent

- Analyze news data (news_agent_output)
- Load and analyze temporal data
 - Load the data from artifacts
- Combined analysis
 - Correlation, timing of news vs stock prices, market reactions to different types of news
- Return summary of analysis to Workflow Agent
 - News, Trends, Combined Insights, Trading Recommendations, Tables

Our Architecture



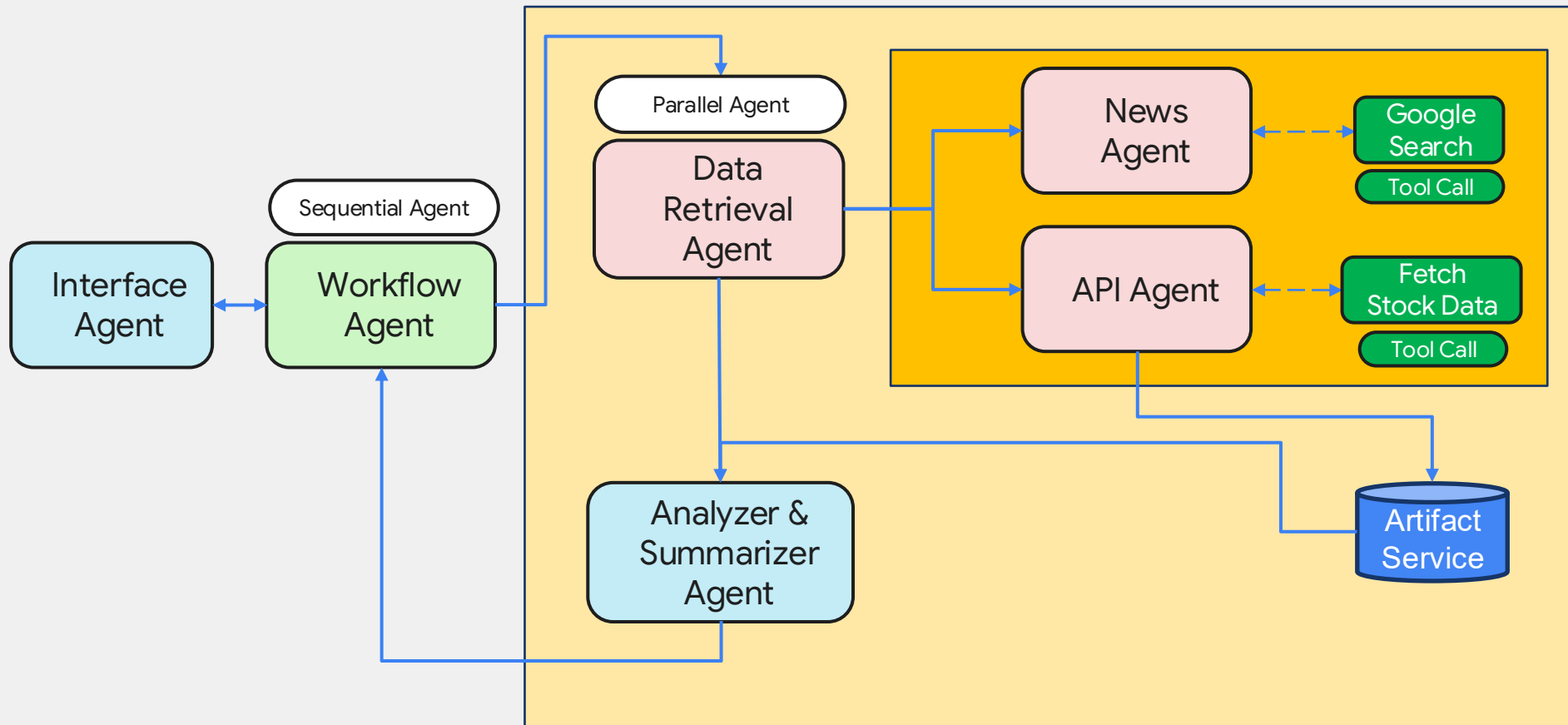
Adapt To Your Use Case

- ADK forces modular architectures
- Plug-and-play
- Update Prompts
- Add / Replace tools & agents

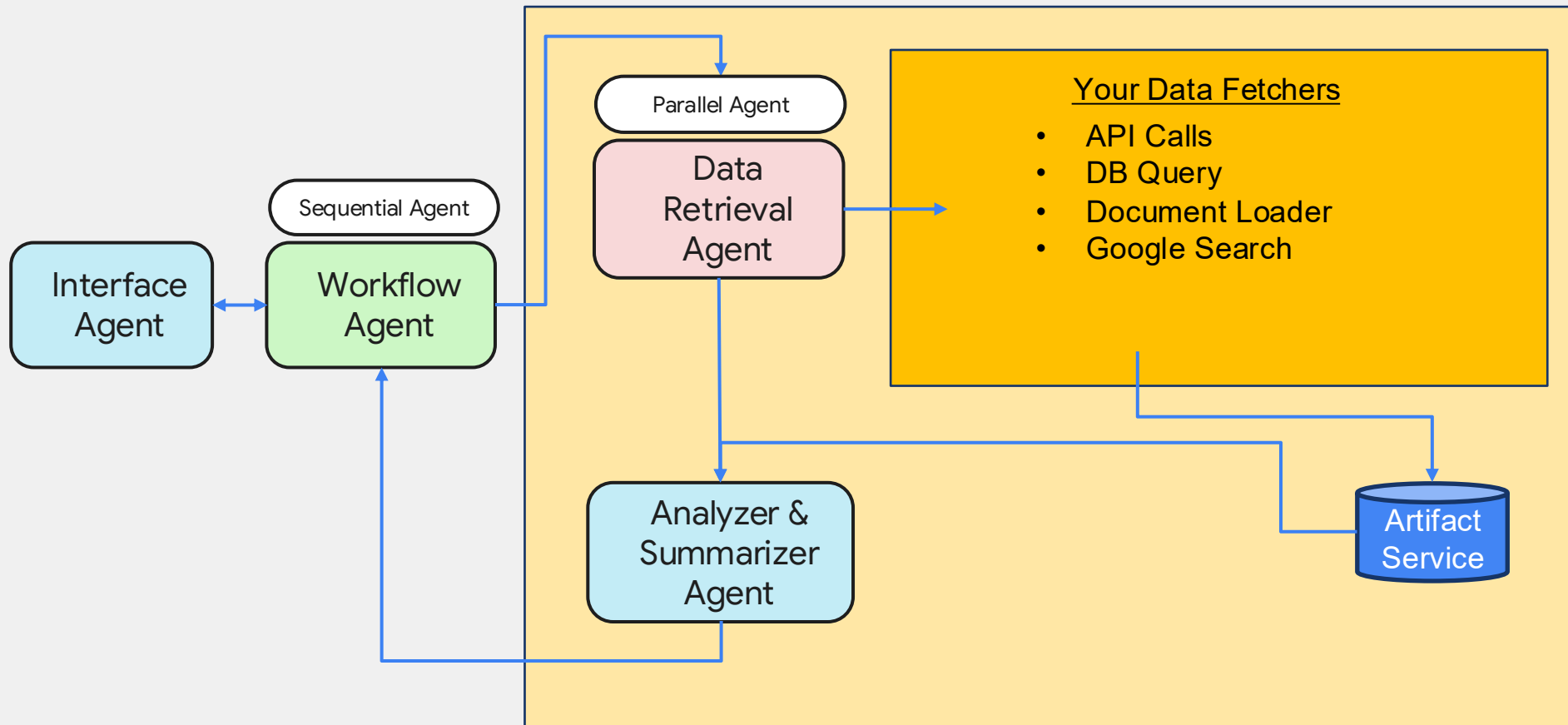
Github repo with code



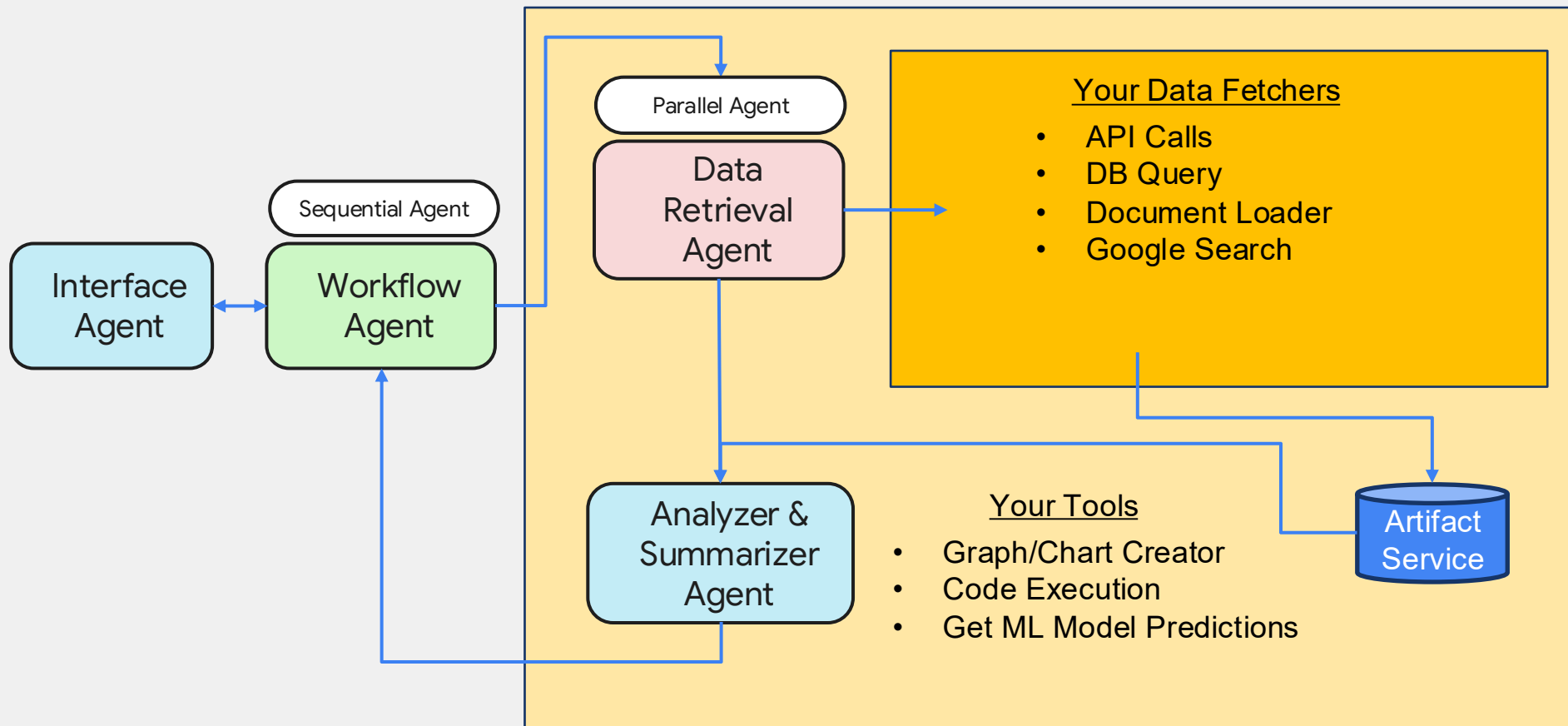
Our Architecture



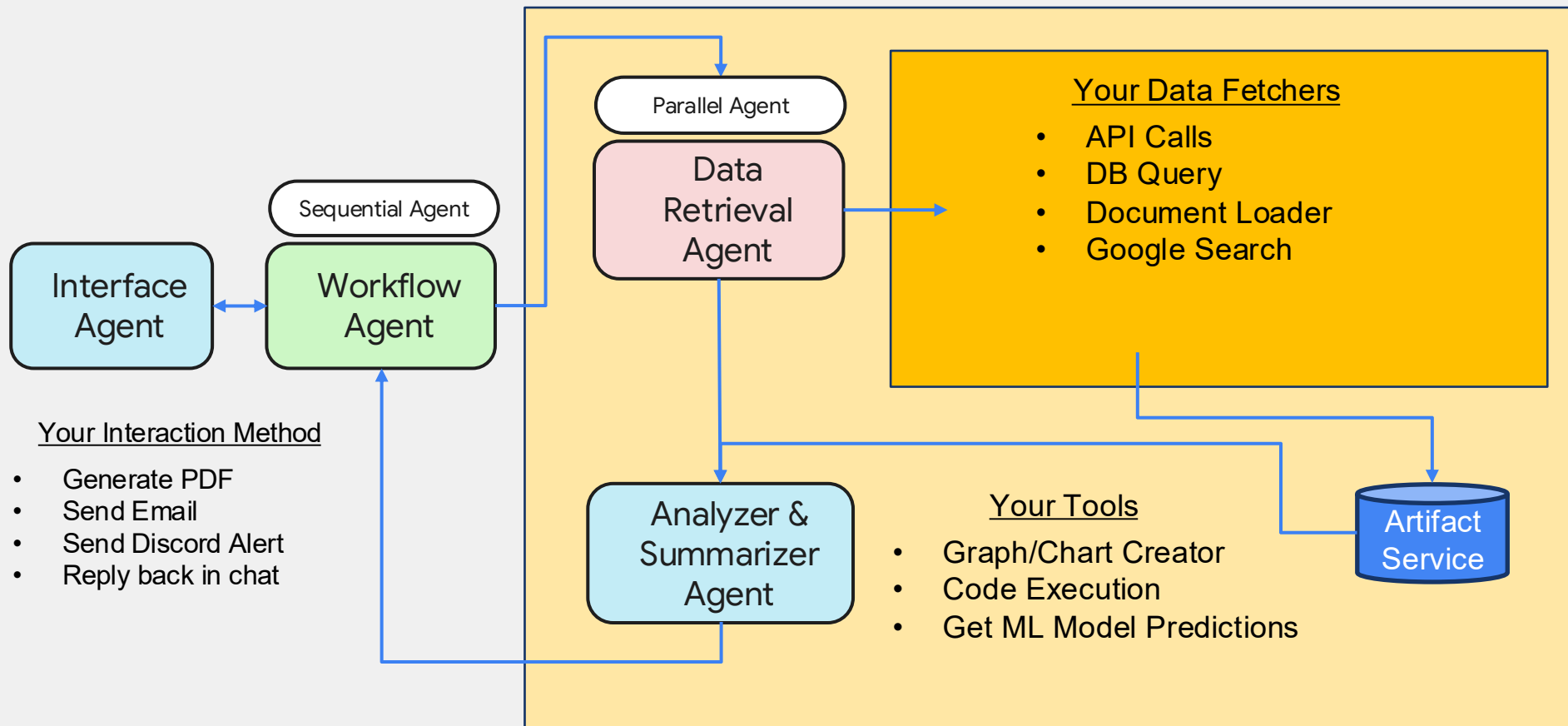
Your Architecture

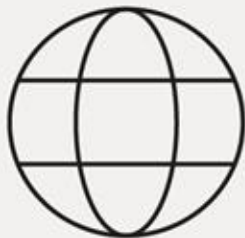


Your Architecture



Your Architecture





Demo Code on Github



github.com/ro1406/adk-report-generator/



Google
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Running Locally



- Local LLMs:

- ☐ **OLLama:** Download and run open-source models locally
- ☐ <https://ollama.com/>



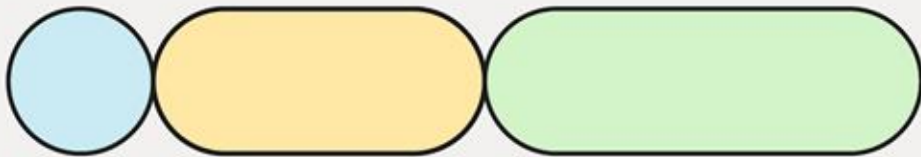
- Free APIs:

- ☐ **Google AI Studio:** Free tier with several models like Gemini 3.5 Flash
- ☐ Also includes “Build Mode” where you build apps with prompts!
- ☐ <https://aistudio.google.com/>



- Run ADK Agents Locally:

- ☐ Just run `adk web` in your terminal



Thank You!

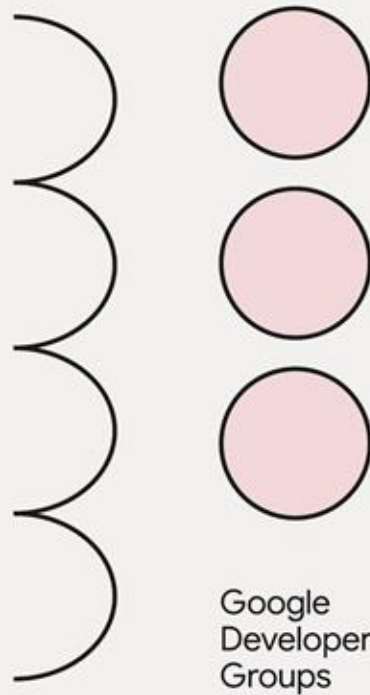


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